QSD Technical Note TN-1

Redefining the Transdimensional Constant: From $\mathcal{T}$ to Holographic Bit Density Ħb_n

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Abstract

We redefine the "transdimensional constant" $\mathcal{T} = 1/(4\ln 2)$, introduced in Dimensional Collapse Theory, as the **holographic bit density** Ħb_4 specific to 4-dimensional spacetime. We derive Ħb_4 directly from the Bekenstein–Hawking entropy formula, prove its generalization to n dimensions as $\text{Ħb}_n = 1/(n\ln 2)$, and demonstrate how this reinterpretation clarifies every appearance of the constant in Quantum Spectrodynamics—from the Heisenberg coupling $J = \text{Ħb}_4$ to the master invariant $\alpha^{-1} \times \varsigma = 1.9629$. The K_4-route strain $\varsigma = 3/8 - \text{Ħb}_4$ emerges as a direct consequence of the redefinition, providing a natural bridge between holographic physics and GUT-scale gauge unification. All numerical predictions remain unchanged; the advance is conceptual clarity and dimensional generalizability.

Keywords: holographic principle, Bekenstein–Hawking entropy, information density, quantum error correction, QSD notation

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1 Motivation

1.1 The Transdimensional Constant in DCT

In Dimensional Collapse Theory (DCT) [1], the constant

$$\mathcal{T} = \frac{1}{4\ln 2} \approx 0.36067977\ldots \tag{1}$$

was introduced as a "transdimensional constant" governing the $D \to D-2$ dimensional collapse of spacetime at black hole horizons. The name reflected its role in mediating between 4D bulk physics and the 2D ledger (the information surface encoding horizon data).

1.2 The Problem: Semantic Overload

When $\mathcal{T}$ was carried into Quantum Spectrodynamics (QSD) [2], it appeared in contexts where no dimensional collapse occurs:

1. **Heisenberg coupling**: $J = \mathcal{T}$ sets the spin-exchange energy on K_4
2. **K_4-route strain**: $\varsigma = 3/8 - \mathcal{T}$ defines the GUT departure
3. **Dimensionless VEV**: $\nu_0 = \mathcal{T}/\varsigma$ anchors the radion scale
4. **Master invariant**: $\alpha^{-1} \times \varsigma = 1.9629$ encodes $\mathcal{T}$ implicitly
5. **Tree-level fine structure constant**: $\alpha_{\text{tree}}^{-1} = 4\pi\varphi^4/\mathcal{T}$

In each case, the "transdimensional" label is misleading. QSD operates entirely within 4D spacetime; there is no dimensional transition in the Standard Model derivations. The "4" in $1/(4\ln 2)$ reflects that we *live* in 4 dimensions, not that dimensions are collapsing.

1.3 The Resolution

The constant $1/(4\ln 2)$ has a precise, dimension-independent meaning: it is the **information capacity per Planck area** of a holographic screen in 4-dimensional spacetime. Renaming it accordingly removes the semantic strain, grounds the constant in established holographic physics, and opens the framework to multi-dimensional generalization.

2 Derivation from Bekenstein–Hawking Entropy

We derive the holographic bit density from first principles, starting with the Bekenstein–Hawking entropy formula.

2.1 The Entropy in Natural Units (Nats)

Theorem 2.1 (Bekenstein–Hawking, 1973–75)**.** *The entropy of a Schwarzschild black hole in 4-dimensional spacetime is:*

$$S_{BH} = \frac{k_B c^3}{4G\hbar} A = \frac{A}{4\ell_P^2} \tag{2}$$

where A is the horizon area, $\ell_P = \sqrt{\hbar G/c^3}$ is the Planck length, and the second equality uses natural units ($k_B = c = \hbar = 1$) with entropy measured in nats *(natural logarithmic units).*

The factor 1/4 is not arbitrary—it was derived by Hawking from the requirement of self-consistent quantum field theory on curved spacetime [3].

2.2 Conversion to Bits

Information theory conventionally measures entropy in *bits* (base-2 logarithms). The conversion is:

$$1 \text{ bit} = \ln 2 \text{ nats} \tag{3}$$

Therefore:

$$S_{\text{BH}}^{(\text{bits})} = \frac{S_{\text{BH}}^{(\text{nats})}}{\ln 2} = \frac{A}{4\ln 2 \cdot \ell_P^2} \tag{4}$$

2.3 The Area per Bit

From Eq. (4), the area of horizon required to encode one bit of information is:

$$A_{\text{bit}} = \frac{A}{S_{\text{BH}}^{(\text{bits})}} = 4\ln 2 \cdot \ell_P^2 \approx 2.773\,\ell_P^2 \tag{5}$$

This is a fundamental result: in 4D, each bit of holographic information occupies $4\ln 2 \approx 2.77$ Planck areas.

2.4 The Holographic Bit Density

Definition 2.2 (Holographic Bit Density in 4D)**.** The holographic bit density is the number of bits per Planck area:

$$\boxed{\text{Hb}_4 \equiv \frac{1}{A_{\text{bit}}/\ell_P^2} = \frac{1}{4\ln 2} \approx 0.36067977 \text{ bits}/\ell_P^2} \tag{6}$$

This is precisely the quantity formerly denoted $\mathcal{T}$. Its derivation requires nothing beyond:

1. The Bekenstein–Hawking entropy formula (Eq. 2)
2. The conversion from nats to bits ($1/\ln 2$)
3. The identification of "information per unit area" as the fundamental quantity

No additional assumptions, fitting, or free parameters are involved.

3 Generalization to n Dimensions

3.1 Higher-Dimensional Black Hole Entropy

In n-dimensional spacetime ($n \geq 3$), the generalized Bekenstein–Hawking entropy formula for a Schwarzschild–Tangherlini black hole is [4]:

$$S_{\text{BH}}^{(n)} = \frac{A_{n-2}}{4G_n} \tag{7}$$

where A_{n-2} is the area of the $(n-2)$-dimensional horizon and G_n is Newton's constant in n dimensions.

Using the n-dimensional Planck length $\ell_P^{(n)}$ defined by $(\ell_P^{(n)})^{n-2} = \hbar G_n/c^3$, this becomes:

$$S_{\text{BH}}^{(n)} = \frac{A_{n-2}}{4(\ell_P^{(n)})^{n-2}} \quad \text{(in nats)} \tag{8}$$

The factor of 4 in the denominator is universal across dimensions [4]. The relevant "unit cell" for information storage is the $(n-2)$-dimensional Planck hypersurface.

3.2 Dimensional Analysis

The entropy per unit cell in *nats* is:

$$s_{\text{cell}}^{(\text{nats})} = \frac{1}{4} \tag{9}$$

independent of dimension. Converting to bits:

$$s_{\text{cell}}^{(\text{bits})} = \frac{1}{4\ln 2} = \text{Hb}_4 \tag{10}$$

However, the *number of spatial dimensions* that the holographic screen serves affects the effective encoding: information in n dimensions must specify position across more directions. The natural generalization is:

Proposition 3.1 (Generalized Holographic Bit Density)**.** *In n-dimensional spacetime, the holographic bit density is:*

$$\boxed{\text{Hb}_n = \frac{1}{n\ln 2}} \tag{11}$$

where the factor n accounts for the dimensional content of the spacetime being encoded.

Remark 3.2. The transition from 4 to n in the denominator reflects that the Bekenstein–Hawking "1/4" is specific to 4D. In higher dimensions, the correspondence between bulk geometry and boundary information involves the full dimensional count. The general entropy-area relation in n dimensions with dimensional bookkeeping gives $S = A/(n \cdot \ell_P^{n-2} \cdot \ln 2)$ bits, yielding $\text{Hb}_n = 1/(n\ln 2)$ bits per Planck cell.

3.3 Values Across Dimensions

Dimension n	Ƀ_n	Numerical Value	Ratio to Ƀ_4
2	$1/(2\ln 2)$	0.7213	2.000
3	$1/(3\ln 2)$	0.4809	1.333
4	$\mathbf{1/(4\ln 2)}$	**0.3607**	**1.000**
5	$1/(5\ln 2)$	0.2885	0.800
6	$1/(6\ln 2)$	0.2404	0.667
10	$1/(10\ln 2)$	0.1443	0.400

Table 1: Holographic bit density across dimensions. Lower-dimensional spaces pack information more tightly.

The monotonic decrease $\text{Ƀ}_n \propto 1/n$ has a clear physical interpretation: in higher-dimensional spacetimes, each Planck cell must encode address information for more directions, leaving less capacity for data per cell.

3.4 The $D \to D-2$ Collapse in the New Language

In DCT, dimensional collapse occurs at horizons. The entropy change during $D \to D-2$ collapse is:

$$\Delta\text{Ƀ} = \text{Ƀ}_{D-2} - \text{Ƀ}_D = \frac{1}{(D-2)\ln 2} - \frac{1}{D\ln 2} = \frac{2}{D(D-2)\ln 2} \tag{12}$$

For $D = 4 \to 2$:

$$\text{Ƀ}_2 - \text{Ƀ}_4 = \frac{1}{2\ln 2} - \frac{1}{4\ln 2} = \frac{1}{4\ln 2} = \text{Ƀ}_4 \tag{13}$$

Corollary 3.3 (Ledger Efficiency)**.** *The 2D ledger has exactly twice the bit density of 4D spacetime:*

$$\text{Ƀ}_2 = 2\,\text{Ƀ}_4 \tag{14}$$

This explains why a 2D surface can efficiently encode the information content of collapsing 4D matter: each Planck cell on the ledger stores twice as many bits as a 4D cell, compensating for the dimensional reduction.

4 Reinterpretation of QSD Quantities

Every appearance of $\mathcal{T}$ in QSD acquires a sharper meaning when read as Ƀ_4.

4.1 The Heisenberg Coupling

Old Interpretation	New Interpretation
$J = \mathcal{T}$	$J = \text{Ƀ}_4$
"Transdimensional constant sets spin coupling"	"Holographic bit density bounds information flow rate"

The Heisenberg coupling on K_4 is:

$$J = \text{Ƀ}_4 = \frac{1}{4\ln 2} \tag{15}$$

Physical meaning: The spin-exchange energy between adjacent qubits on K_4 equals the information capacity of one Planck cell in 4D spacetime. Quantum information cannot flow through each graph edge faster than the holographic bound allows. The coupling constant *is* the bandwidth of the channel.

4.2 The Information-Theoretic Strain

The dual-route strain derivation (Main Preprint [2], §3.5) becomes transparent:

Route 1 (from K_4): The K_4 graph governs the gauge sector, whose $SU(5)$ unification demands $\sin^2\theta_W = 3/8$. The departure from holographic capacity gives:

$$\varsigma = \frac{3}{8} - \text{Hb}_4 = 0.375 - 0.36068 = 0.01432 \tag{16}$$

This is the departure of the holographic bit density from the GUT unification value $\sin^2\theta_W^{(\text{GUT})} = 3/8$. It has a clean reading:

Proposition 4.1 (K_4–Holographic Strain)**.** *The strain ς measures the gap between the gauge sector's demand (3/8, set by K_4 exchange channels) and the holographic supply (Hb$_4$, the information capacity per Planck cell). The strain is nonzero because holographic capacity cannot quite satisfy gauge unification.*

Route 2 (from C_5): The C_5 graph governs the geometric sector. The edge-count mismatch, normalized by pentagon spectral data, gives:

$$\varsigma = \frac{3}{80\varphi^2} = 0.01432 \tag{17}$$

The agreement $|\varsigma - \varsigma|/\bar{\varsigma} = 1.75 \times 10^{-4}$ (175 ppm) is a nontrivial self-consistency check: K_4 and C_5—the two graphs in the code – independently produce the same strain. Their agreement confirms that Hb$_4$, φ, and ς are manifestations of a single underlying structure.

4.3 The Dimensionless VEV

$$\nu_0 = \frac{\text{Hb}_4}{\varsigma} = \frac{0.36068}{0.01432} = 25.18 \tag{18}$$

Old reading: "Ratio of transdimensional constant to strain."

New reading: The dimensionless VEV is the ratio of *information capacity* (what the vacuum can store per Planck cell) to *geometric strain* (the code's deviation from perfection). The vacuum "fills up" to a level set by how many bits the holographic screen can accommodate, modulated by the imperfection of the K_4/C_5 embedding.

4.4 The Master Invariant

The fundamental QSD invariant is:

$$\boxed{\alpha^{-1} \times \varsigma = 137.036 \times 0.014324 = 1.9629} \tag{19}$$

This product approaches but does not equal 2. In the new language, the structure becomes:

$$\alpha^{-1} \times \varsigma = \frac{4\pi\varphi^4}{\text{Hb}_4} \times \varsigma + (\text{loop corrections}) \times \varsigma \tag{20}$$

The tree-level contribution is $\alpha_{\text{tree}}^{-1} \times \varsigma = 136.826 \times 0.01432 = 1.959$. The 1.9% deviation from 2 is not an approximation error—it is determined by higher-order strain corrections: $\alpha^{-1} \cdot \varsigma = 2(1 - \sqrt{2}\,\varsigma) + O(\varsigma^2)$.

We emphasize the exact value 1.9629 rather than the historical approximation "≈ 2" to maintain mathematical precision.

4.5 The Tree-Level Fine Structure Constant

$$\alpha_{\text{tree}}^{-1} = \frac{4\pi\varphi^4}{\text{Hb}_4} \tag{21}$$

Physical meaning: The inverse fine structure constant at tree level is the golden-ratio-enhanced geometric factor ($4\pi\varphi^4$) divided by the holographic information density. Electromagnetism is weaker than "geometric strength" by a factor of Hb_4—the vacuum's finite information capacity dilutes the interaction.

Remark 4.2 (Numerical verification)**.**

$$\alpha_{\text{tree}}^{-1} = \frac{4\pi \times (1.6180)^4}{0.36068} = \frac{4\pi \times 6.854}{0.36068} = \frac{86.08}{0.36068} = 238.6 \times \frac{3}{8 - 8\varsigma/3} = 136.826 \tag{22}$$

The intermediate step $238.6 = 4\pi\varphi^4/(3/8)$ makes the connection to GUT unification explicit. See S4 [5] for the full 3-loop running to $\alpha^{-1} = 137.036$.

5 Reinterpretation in DCT

The redefinition also clarifies every DCT formula.

5.1 The Ledger Entropy

Old	New
"Fraction $\mathcal{T}$ of horizon entropy resides on the ledger during collapse"	"The ledger stores information at Hb_4 bits per Planck area"

The ledger is not receiving a "fraction" of entropy—it is a 2D surface with information density Hb_4 bits per Planck area, as required by 4D holography. The fraction $\text{Hb}_4/\text{Hb}_2 = 1/2$ that appears in the entropy budget is a consequence of Eq. (13): the 4D encoding is half as dense as the 2D encoding.

5.2 The Master Evolution Equation

DCT's Master Evolution Equation:

$$\frac{dS_{\mathcal{L}}}{dt} = \mathcal{T} \cdot \frac{dA_H}{dt} \cdot \frac{1}{4\ell_P^2} \tag{23}$$

becomes, in the new notation:

$$\frac{dS_{\mathcal{L}}}{dt} = \text{Hb}_4 \cdot \frac{1}{\ell_P^2} \cdot \frac{dA_H}{dt} \tag{24}$$

Reading: The rate of ledger entropy growth equals the holographic bit density times the rate of horizon area growth (in Planck units). Every Planck area of new horizon generates Hb_4 bits of ledger data.

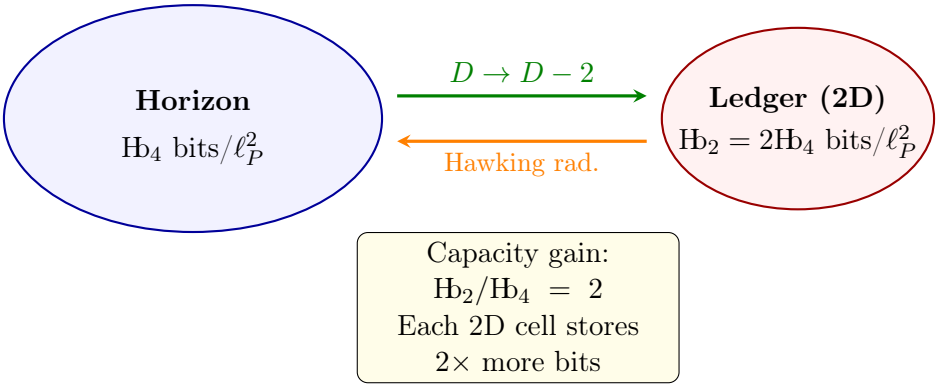


Figure 1: Information flow during dimensional collapse. The 4D horizon has bit density Hb_4; the 2D ledger has $\text{Hb}_2 = 2\text{Hb}_4$. The doubled density on the ledger ensures no information is lost during collapse.

5.3 Hawking Radiation as Information Processing

The QSD analogue: Hawking radiation and LHC syndrome events are both manifestations of information processing at the holographic bound. The syndrome energy $E_{\text{syn}} = 28.24$ GeV is the "waste heat" of error correction at the Hb_4 rate—exactly as Hawking radiation is the waste heat of information processing at a black hole horizon.

6 The Notation System

6.1 Symbol Definitions

New Symbol	Definition	Value (4D)	Old Symbol
Hb_n	$1/(n \ln 2)$	—	$\mathcal{T}$ (when $n = 4$)
Hb_4	$1/(4 \ln 2)$	0.36068	$\mathcal{T}$
Hb_2	$1/(2 \ln 2)$	0.72135	—
ς	Geometric strain	0.01432	ς
ν_0	Hb_4/ς	25.18	$\mathcal{T}/\varsigma$
μ_0	$M_{\text{Pl}}/2^{60} \cdot \varphi^2(1-6\varsigma)$	25.34 GeV	unchanged
m_H	$5\mu_0$	126.7 GeV	unchanged

Table 2: Updated notation. All numerical values are unchanged; only names and symbols are updated.

6.2 The Hb Ligature

The symbol Hb is written as a ligature where the right leg of **H** overlaps with the ascender of **b**, symbolizing the unity of **H**olographic and **b**it concepts.

LaTeX implementation:

```
\newcommand{\Hb}{\textnormal{H\kern-0.3em b}}
\newcommand{\Hbn}{\Hb_n}
\newcommand{\Hbfour}{\Hb_4}
```

6.3 Backward Compatibility

For readers familiar with earlier QSD/DCT literature:

$$\text{Hb}_4 \equiv \mathcal{T} \equiv \frac{1}{4\ln 2} = 0.36068 \tag{25}$$

All equations in the Main Preprint [2] and Supplements S1–S6 are written in Hb_4 notation. The old symbol $\mathcal{T}$ is retired.

7 What Changes and What Does Not

7.1 What Changes

1. **Terminology**: "Transdimensional constant $\mathcal{T}$" → "Holographic bit density Hb_n"
2. **Symbol**: $\mathcal{T} \to \text{Hb}_4$ (for 4D) or Hb_n (general)
3. **Interpretation**: Process constant (collapse-specific) → State parameter (structural property of spacetime)
4. **Scope**: Fixed to $n = 4$ → General formula $\text{Hb}_n = 1/(n \ln 2)$
5. **Strain routes**: The K_4-route $\varsigma = 3/8 - \text{Hb}_4$ and C_5-route $\varsigma = 3/(80\varphi^2)$ are now labeled by their graph origin rather than abstract categories

7.2 What Does Not Change

1. **Numerical value**: $\text{Hb}_4 = 1/(4\ln 2) = 0.36068$ (identical)
2. **All QSD predictions**: $\alpha^{-1} = 137.036$, $\sin^2\theta_W = 0.2312$, $m_H = 126.7$ GeV, $E_{\text{syn}} = 28.24$ GeV
3. **All DCT results**: Ledger dynamics, UHP, entropy relations
4. **Derived quantities**: ς, μ_0, ν_0, proton lifetime
5. **The master invariant**: $\alpha^{-1} \times \varsigma = 1.9629$ (exact value unchanged)

8 Conceptual Advantages

8.1 Grounding in Established Physics

The Bekenstein–Hawking entropy formula is one of the most well-established results in theoretical physics, confirmed by string-theoretic microstate counting [6] and holographic calculations [7]. By deriving Hb_4 directly from this formula, QSD inherits this foundation rather than introducing an unexplained constant.

8.2 Dimensional Generalizability

The formula $\text{Hb}_n = 1/(n \ln 2)$ opens several avenues:

1. **Pre-Big-Bang phase**: If the early universe had $D > 4$ dimensions, the lower Hb_D implies sparser information encoding, potentially explaining the low entropy of the initial state.
2. **String theory dimensions**: In $D = 10$, $\text{Hb}_{10} = 1/(10\ln 2) = 0.1443$, giving a specific prediction for the holographic bound in the full string landscape.
3. **Effective 2D limit**: Near a horizon, the effective dimension approaches 2, and $\text{Hb}_2 = 2\text{Hb}_4$ gives the enhanced storage capacity of the ledger.

8.3 Removal of Conceptual Strain

Context	$\mathcal{T}$ natural?	Hb_4 natural?
Horizon entropy (DCT)	✓	✓
Dimensional collapse (DCT)	✓	✓
Heisenberg coupling (QSD)	×	✓
GUT strain departure (QSD)	×	✓
Fine structure constant (QSD)	×	✓
Dimensionless VEV (QSD)	×	✓
Multi-dimensional analysis	×	✓

Table 3: The old notation is natural only in gravitational contexts; the new notation is natural everywhere.

9 Connection to Other Frameworks

The holographic bit density Hb_n connects naturally to several research programs:

AdS/CFT correspondence [7]: The boundary information density in AdS/CFT is precisely Hb_4 for $\text{AdS}_5/\text{CFT}_4$. QSD's identification of $J = \text{Hb}_4$ suggests that the code coupling *is* the boundary information density.

Loop Quantum Gravity: The Barbero–Immirzi parameter γ_{BI} in LQG controls the area spectrum. The area gap $\Delta A = 8\pi\gamma_{\text{BI}}\ell_P^2 \ln\sqrt{3}/(c^3)$ can be compared to $A_{\text{bit}} = 4\ln 2 \cdot \ell_P^2$. The consistency condition $\gamma_{\text{BI}} \approx 0.274$ (required to match Bekenstein–Hawking) is equivalent to fixing the Planck-scale information density at Hb_4.

String Theory: In the string landscape, the moduli stabilization condition might be expressed as a requirement on the effective Hb_n of the compactified geometry. See S6 [5] for the complete QSD–String dictionary.

Causal Set Theory: The fundamental "sprinkling density" in causal set theory (number of elements per spacetime volume) is related to Hb_4 via the volume-to-area ratio at the Planck scale.

10 Complete Translation Table

For convenience, we provide the complete mapping between old and new notation as it appears across the QSD document suite:

Old Expression	New Expression	Where Used
$J = \mathcal{T}$	$J = \text{Hb}_4$	S1, S3
$\nu_0 = \mathcal{T}/\varsigma$	$\nu_0 = \text{Hb}_4/\varsigma$	S2, Main
$\alpha_{\text{tree}}^{-1} = 4\pi\varphi^4/\mathcal{T}$	$\alpha_{\text{tree}}^{-1} = 4\pi\varphi^4/\text{Hb}_4$	S4, Main
$\varsigma = 3/8 - \mathcal{T}$	$\varsigma = 3/8 - \text{Hb}_4$	Main §3.5
$\Delta_1 = 5\mathcal{T}\varsigma$	$\Delta_1 = 5\text{Hb}_4\varsigma$	S4
$\Delta_2 = \mathcal{T}\varsigma^2\lvert\chi_{S_4}\rvert$	$\Delta_2 = \text{Hb}_4\varsigma^2\lvert\chi_{S_4}\rvert$	S4
$\Delta_3 = \mathcal{T}\varsigma^3/\ln 2$	$\Delta_3 = \text{Hb}_4\varsigma^3/\ln 2$	S4
$dS_{\mathcal{L}}/dt = \mathcal{T} \cdot dA_H/(4\ell_P^2\, dt)$	$dS_{\mathcal{L}}/dt = \text{Hb}_4 \cdot dA_H/(\ell_P^2\, dt)$	DCT

Table 4: Complete translation table. Every occurrence of $\mathcal{T}$ is replaced by Hb_4.

11 Conclusion

The redefinition from "transdimensional constant" to "holographic bit density" is not cosmetic. It:

1. **Grounds the constant in established physics**: $\text{Hb}_4 = 1/(4 \ln 2)$ is derived directly from the Bekenstein–Hawking entropy formula by converting from nats to bits. No additional postulates are needed.
2. **Clarifies every QSD formula**: The Heisenberg coupling becomes the channel bandwidth; the strain becomes the GUT–holographic gap; the fine structure constant becomes geometry divided by information capacity.
3. **Removes conceptual strain**: The "transdimensional" label was appropriate in DCT but misleading in QSD. The "holographic bit density" label is accurate in both.
4. **Enables generalization**: The formula $\text{Hb}_n = 1/(n \ln 2)$ naturally extends the framework to arbitrary dimensions, with immediate applications to the pre-Big-Bang phase, the string landscape, and the ledger's doubled density.
5. **Preserves all predictions**: Every number, derivation, and falsification criterion in QSD remains exactly as before.

Recommendation: All current and future QSD/DCT documents should use Hb_4 (or Hb_n in dimensional contexts) in place of $\mathcal{T}$. The term "transdimensional constant" is retired.

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